

HAOYU GAO

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Google Scholar • ORCID • GitHub • Homepage

RESEARCH INTEREST

My research spans empirical and automated software engineering, with a focus on how developers create, maintain, and use documentation. I study documentation as a socio-technical artefact, combining repository mining, qualitative methods, and machine learning to analyse both its technical content and the human practices around it. My current work examines how development knowledge is communicated to AI agents. I am particularly interested in agent configuration files as a medium for encoding developer intent, context, and process knowledge to support effective human–AI collaboration.

EDUCATION

The University of Melbourne

Jan 2023 - Nov 2026 (Expected)

Doctor of Philosophy - Engineering and IT

Supervisors: A/Prof. Christoph Treude and Dr. Mansooreh Zahedi

Research Topic: *Understanding and Maintaining Documentation in Fast-Evolving Software Ecosystems.*

The University of Melbourne

Feb 2021 - Dec 2022

Master of Information Technology

WAM: 86.7/100

Fuzhou University

Sep 2016 - July 2020

Bachelor of Science - Mathematics and Applied Mathematics

WAM: 86.6/100

PUBLICATIONS

[TSE'25] **H. Gao**, C. Treude, and M. Zahedi. Adapting Installation Instructions in Rapidly Evolving Software Ecosystems: In *IEEE Transactions on Software Engineering*, 2025. **(core A*)**

[FSE'23] **H. Gao**, C. Treude, and M. Zahedi. “Evaluating Transfer Learning for Simplifying GitHub READMEs”. In *ESEC/FSE'23: Proceedings of the Joint European Software Engineering Conference and Symposium on the Foundations of Software Engineering, 2023*. **(core A*)**

[EMSE'26] **H. Gao**, M. Zahedi, W. Jiang, H. Lin, J. Davis, and C. Treude. AI Failures in the Eyes of the Downstream Developer: A First Look at Concerns, Practices, and Challenges. *Springer Nature Empirical Software Engineering*, 2026 **(core A)**

[ESEM'24] **H. Gao**, M. Zahedi, C. Treude, S. Rosenstock, and M. Cheong. “Documenting Ethical Considerations in Open Source AI Models”. In *ESEM'24: 18th International Symposium on Empirical Software Engineering and Measurement*, 2024. **(core A)**

[ACL'25 findings] H. Lin, C. Liu, **H. Gao**, P. Thongtanunam, and C. Treude. CodeReviewQA: The Code Review Comprehension Assessment for Large Language Models. In *Findings of ACL'25: The 63rd Annual Meeting of the Association for Computational Linguistics* **(core A*)**

[ICSME'25] P. Banyongrakul, M. Zahedi, P. Thongtanunam, C. Treude, and **H. Gao**. From Release to Adoption: Challenges in Reusing Pre-trained AI Models for Downstream Developers. In *ICSME'25: 41st International Conference on Software Maintenance and Evolution* **(core A)**

[MSR'26 short] **H. Gao**, P. Banyongrakul, H. Guan, M. Zahedi, C. Treude, “On Autopilot? An Empirical Study of Human-AI Teaming and Review Practices in Open Source”. *Short Paper in MSR'26: 23rd International Conference on Mining Software Repositories*, 2026, **(core A)**

H. Gao, H. Lin, C. Treude, G. Gay, and M. Zahedi. Does My README File Need To Be Updated? Exploring LLM-Based README Maintenance. **(Major revision under TSE)**

H. Lin, C. Liu, **H. Gao**, P. Thongtanunam, and C. Treude. Fine-grained Approaches for Confidence Calibration of LLMs in Automated Code Revision. (**Major revision under TSE**)

P. Banyongrakkul, M. Zahedi, C. Treude, **H. Gao** and P. Thongtanunam. When AI Models Become Dependencies: Studying the Evolution of Pre-Trained Model Reuse in Downstream Software Systems. (**Major revision under TSE**)

H. Gao, J. Lulla, H. Lin, S. Baltes, C. Treude, M. Zahedi. From Registry to Repository: How AI Agent Skills Are Written, Adapted, and Maintained. (**Under submission**)

L. Salerno, **H. Gao**, G. Gay, A. Serebrenik, and P. Leitner. Let the Agent Do It? How Software Practitioners Understand and Make Permission Decisions in Agentic AI Assistants. (**Under submission**)

EXPERIENCE

Visiting Postgraduate Research Student, Singapore Management University *Sep 2025- Feb 2026*

Host Professors: A/Prof. Christoph Treude

Teaching Assistant, The University of Melbourne

Oct 2023- Present

Tutored Courses:

- COMP90041 Programming and Software Development
- SWEN90017 Masters Advanced Software Project

Marked Courses:

- COMP90041 Programming and Software Development
- SWEN90016 Software Processes and Management
- COMP90049 Introduction to Machine Learning
- COMP90051 Statistical Machine Learning

Skills: Java Programming Language, Object-Oriented Programming, Software Development Processes, Agile Development, Machine Learning, Assignment Design, University Teaching.

Research Assistant, The University of Melbourne

Dec 2023- May 2024

Skills: Mining software repositories, Qualitative analysis, Ethics in Machine Learning

Machine Learning Internship, Haier Group Corporation

Oct 2021- Feb 2022

Skills: Machine Learning, Data analysis

SERVICES AND ACTIVITIES

Journal Review

- Empirical Software Engineering Journal.
- Automated Software Engineering Journal.

Conference Review

- PC member for ICSME'26 Demo and Data Showcase Track.
- PC member for ICSE'26 Artifact Evaluation Track.
- Junior PC member for MSR'25.

Sub-reviewer

TOSEM, ICSE'25, ESEM'24, APSEC'24.

Student Volunteer

Event: ICSE 2023 - International Conference on Software Engineering

Skills: registration assistance and presentation session support.

(CO)-SUPERVISED STUDENTS

Hao Guan - Master of Software Engineering, The University of Melbourne

SKILLS

Programming

Python, Java, JavaScript, R.

Research

Data mining, Interview-based study, Qualitative analysis, Machine learning, HPC cluster.

Languages

Mandarin (Native), English (Full professional proficiency).

AWARDS AND SCHOLARSHIPS

FEIT PhD Writeup Award, The University of Melbourne

Aug 2026

Funded by the faculty and school with A\$3,500.

FEIT Conference Travel Grant, The University of Melbourne

Feb 2026

Funded by the faculty and school for conference traveling with A\$4,000.

2022 Dean's Honours List, The University of Melbourne

July 2023

The Dean's Honours List recognises outstanding academic performance in the academic year, representing the top five per cent of students enrolled in an engineering or information technology master by coursework program.

Melbourne Research Scholarship, The University of Melbourne

Jan 2023

This scholarship is rewarded to high-achieving students participating in research activities.

Melbourne Graduate Scholarship, The University of Melbourne

Jan 2022

This scholarship is rewarded to coursework students in recognition of their academic achievements.